Nothing Here

Quantum Algorithms Using Clebsch-Gordan Transforms

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funded by



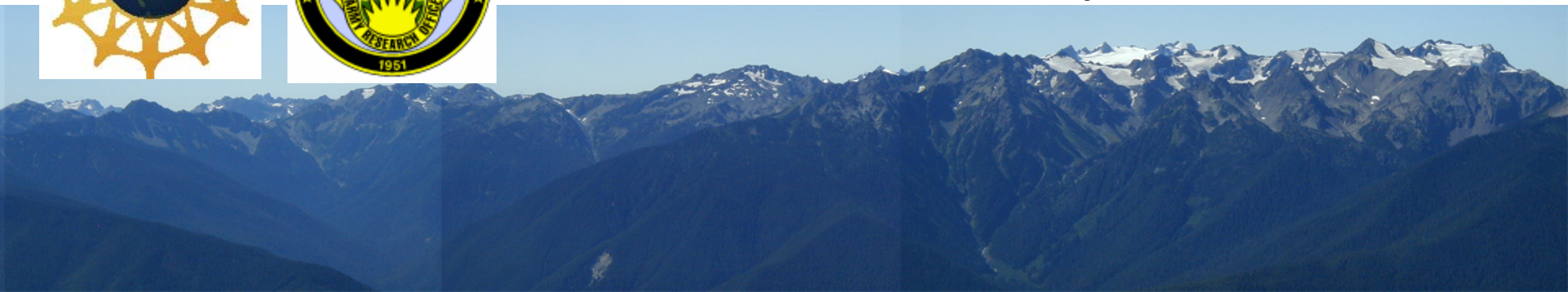








"Que Dubayah"



A Brief History of Hiring in Quantum Computing











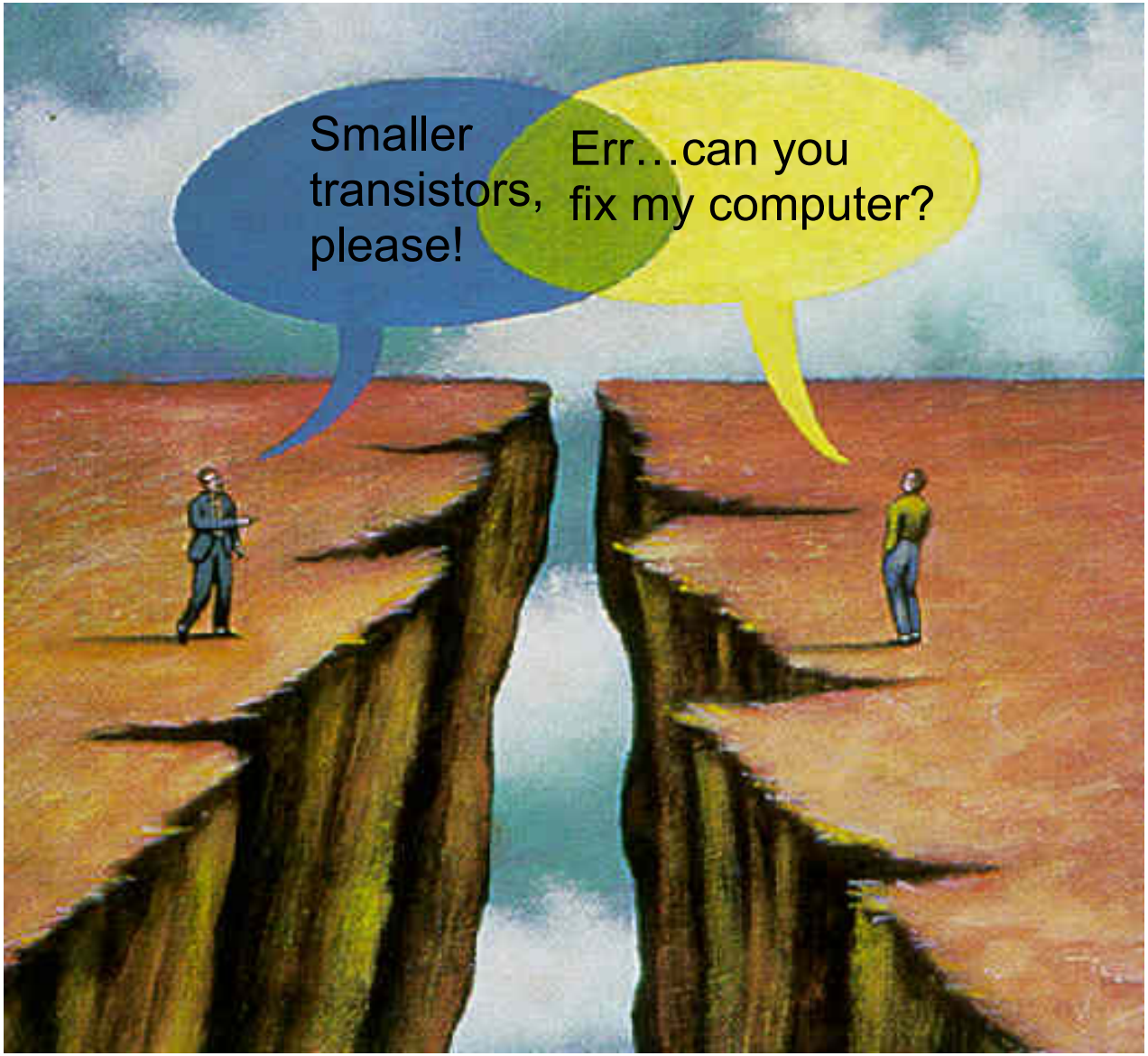








Physics and Computer Science



They're Scared?

Email I received from Wim van Dam about FOCS 2006:

Leslie Valiant in his great talk on 'accidental algorithms': "We have all these complexity classes, and we don't know at all if they are different... It would be a nightmare if they turn out to be all the same, such that NC2=P=P^#P and so on." <comment from the audience about if this would really be so bad> "Yes it would be a nightmare, but the *worst* possible nightmare would be if NC=P=P^#P and if it was a *physicist* who proved it."



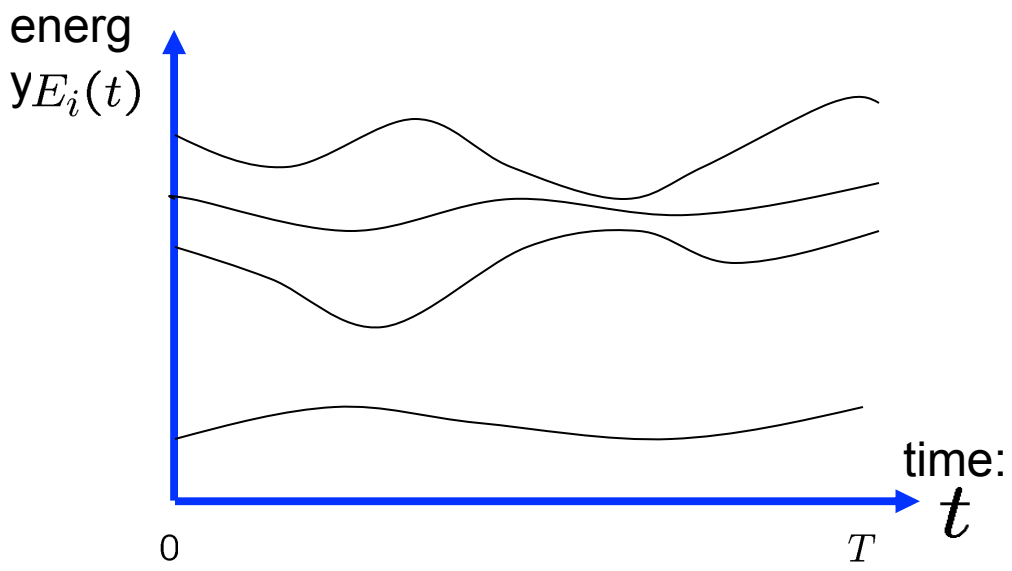
They're Not Scared?

Aaronson

Q: Does quantum computing really belong in CS departments, as opposed to physics departments?

A: It belongs if we want it to belong! In my experience, the *physicists* have a bigger hurdle than the computer scientists in getting started with quantum computing research. All we need to do is ask themselves: "what happens if we generalize probability theory to allow minus signs, and base it on the L2 norm instead of the L1 norm?" From then on it's just the concepts we know and love: states, transformations, recursion, reductions, universality, asymptotic efficiency, and so on. Physicists, by contrast, have to learn most of this stuff for the first time. It's been a great personal pleasure to watch physicists who once suspected that CS was devoid of intellectual content, struggle with that content while trying to learn quantum computing!

Can Physicist Contribute Quantum Algorithms?



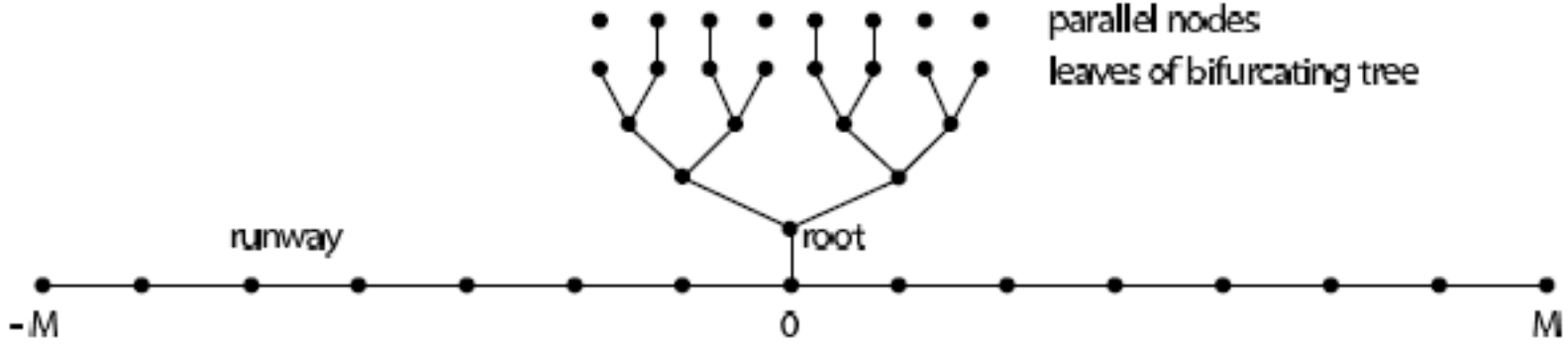


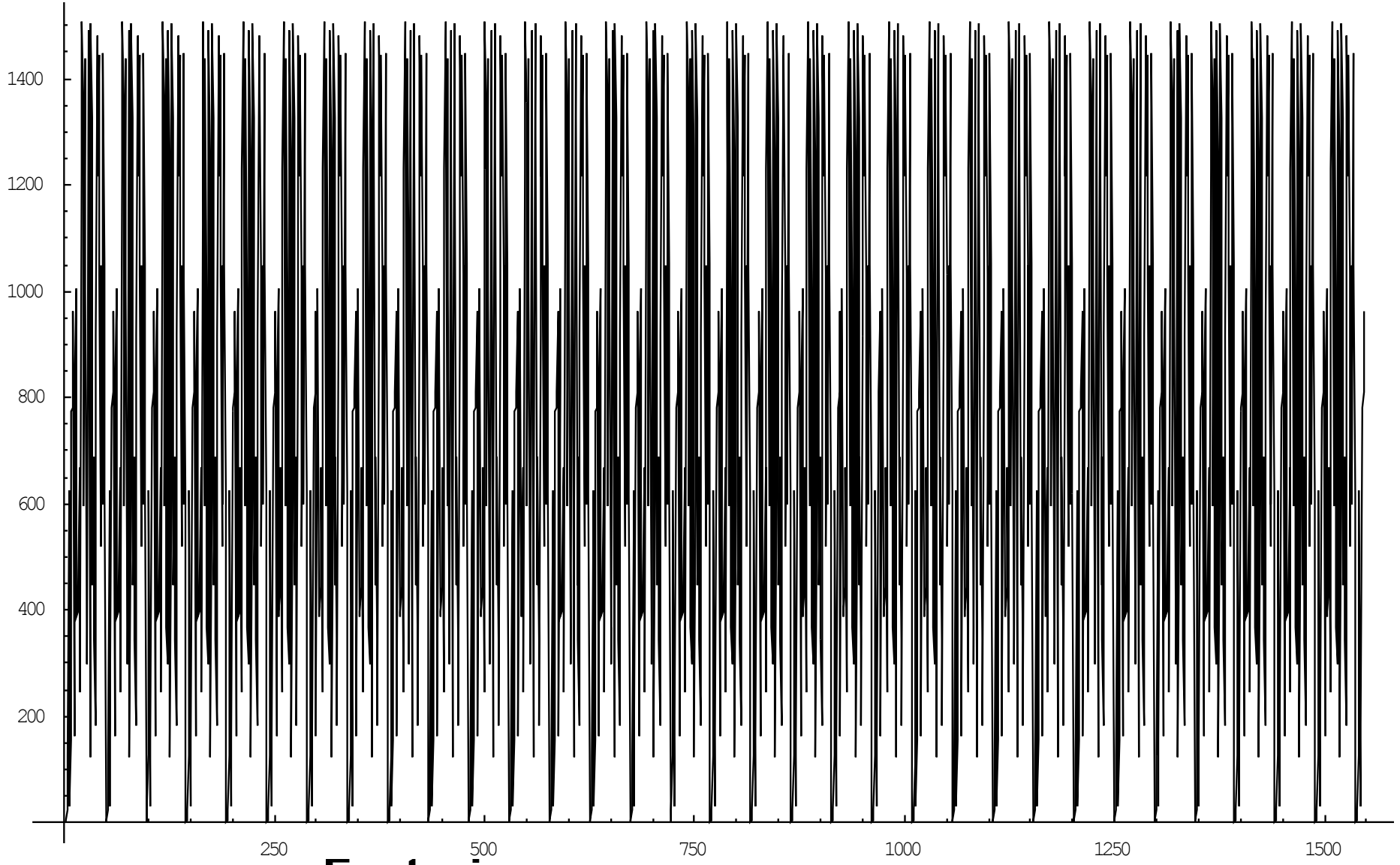
Figure 2: The full Hamiltonian $H_O + H_D$

Bottom fig. from quantph/0702144, Farhi, Goldstone, Guttman

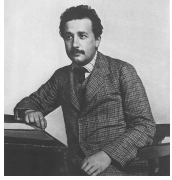
Shor's Algorithm



N=1547, r=5, Period = 48. 48+1=49 GCD(1547,49)=7



Period Finding

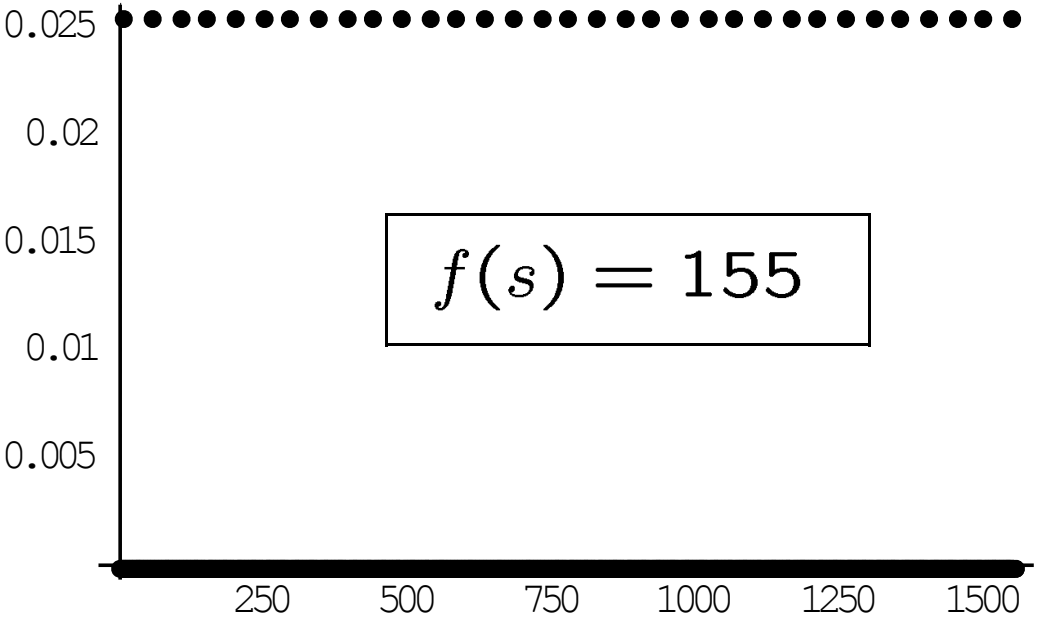


Why hard?

$$f(s) = r^s \bmod N$$



$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$



$$f(s) = 155$$

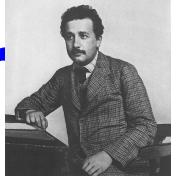
Measure second register

First register is (nearly) symmetric under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^\dagger \approx \rho$$

The First Basis Change We Learn In Quantum Mechanics

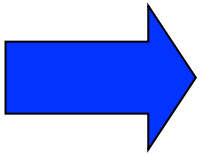


$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$

symmetry under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^{\dagger} = \rho$$



change to momentum basis!

$$|p\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega_N^{ixp} |x\rangle$$

$$\omega_N = \exp\left(\frac{2\pi i}{N}\right)$$

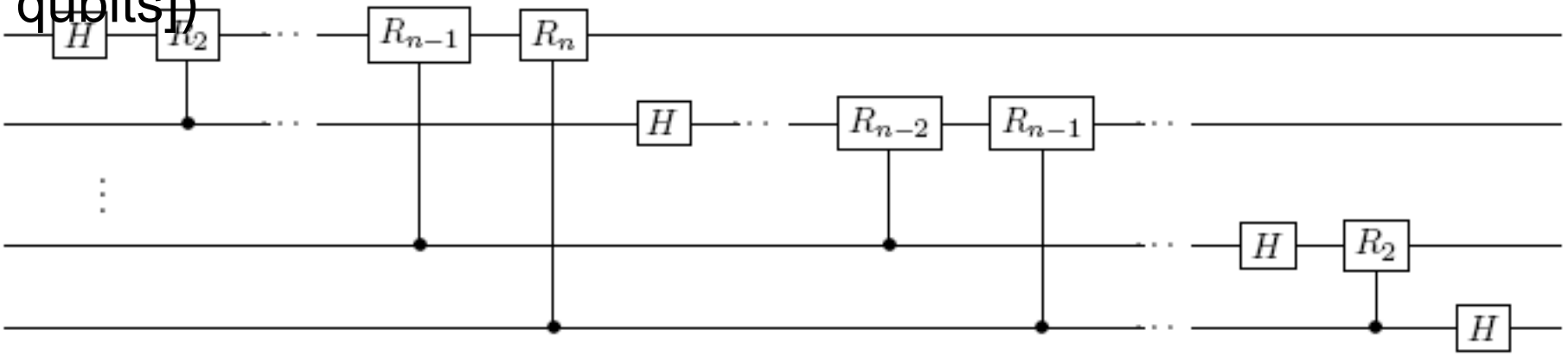
Perform a Fourier Transform!

Quantum Fourier Transform



Efficient quantum circuit for quantum Fourier transform:

[Efficient = O(poly[# qubits])



Factoring



Period Finding

Symmetry





Change Basis

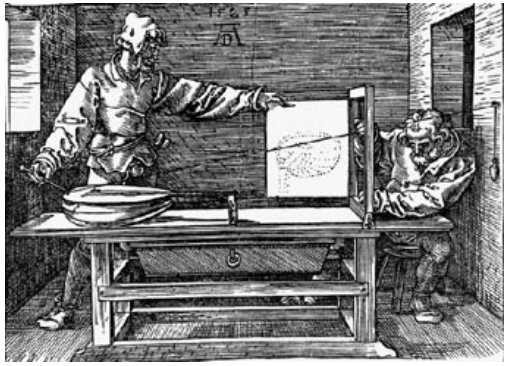
Efficient Circuit



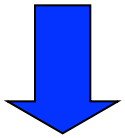


Shor's Algorithm

Moral

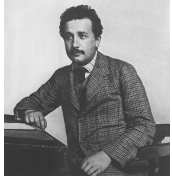


Symmetries



Quantum algorithms

The Second Basis Change We Learn In Quantum Mechanics?



The Second Basis Change We Learn In Quantum Mechanics:

Clebsch-Gordan Transform

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

Triplet:

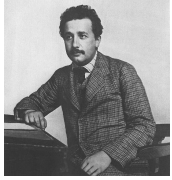
$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

What Use?

Collective Unitaries



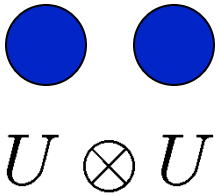
$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

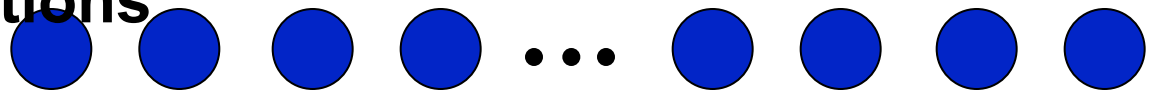
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

$$\left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & U \end{array}\right]$$

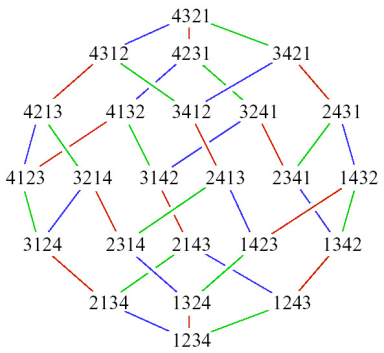


$U \otimes U$

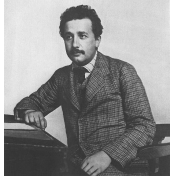
Symmetry Under n Collective Unitary Rotations



$U^{\otimes n}$



The Symmetric Group



Transform Under SWAP

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle\otimes|\downarrow\rangle - |\downarrow\rangle\otimes|\uparrow\rangle) \qquad -1$$

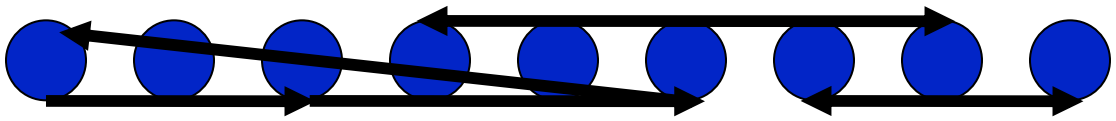
Triplet:

$$|S=1,m=1\rangle = |\uparrow\rangle\otimes|\uparrow\rangle$$

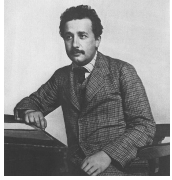
$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle\otimes|\downarrow\rangle + |\downarrow\rangle\otimes|\uparrow\rangle) \qquad +1$$

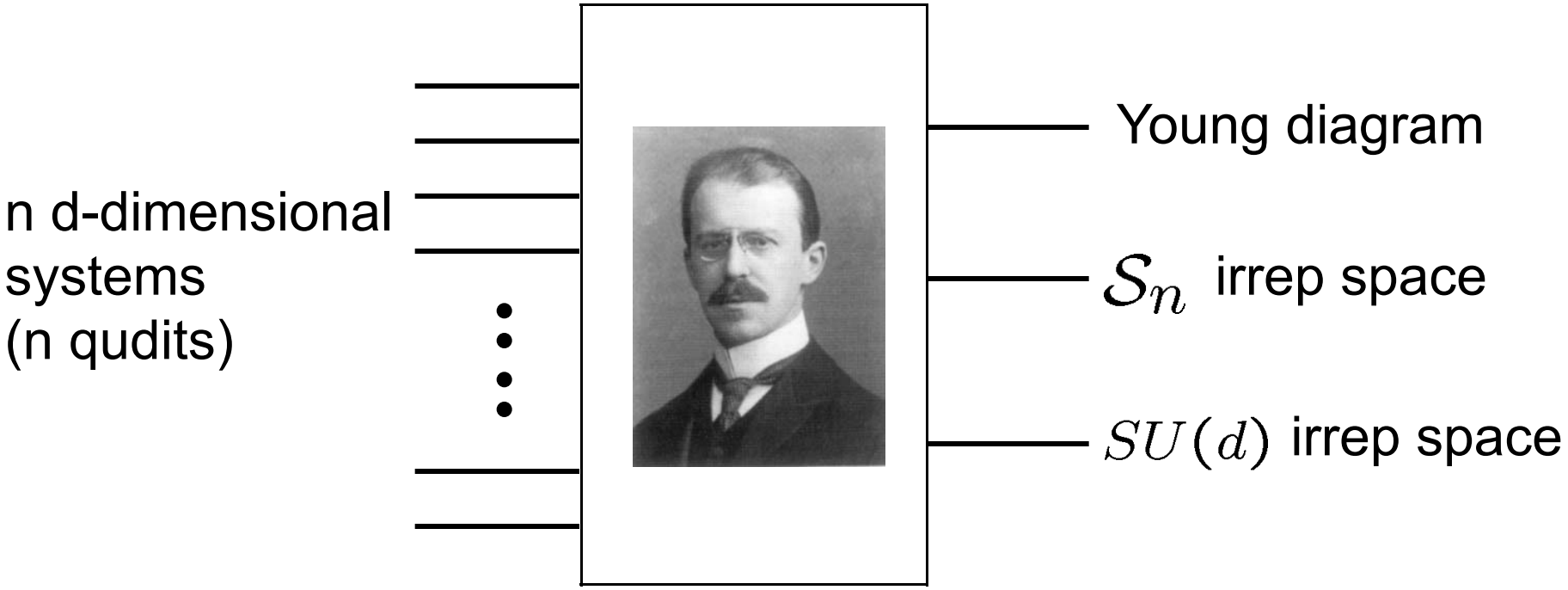
$$|S=1,m=-1\rangle = |\downarrow\rangle\otimes|\downarrow\rangle$$

Symmetry Under Permutation of n Subsystems



Quantum Schur Transform

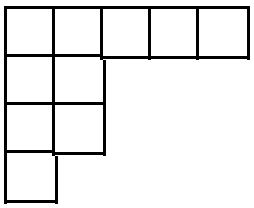




Schur duality

$$[\mathbb{C}^d]^{\otimes n} \overbrace{\cong}^{\text{Schur}} \bigoplus_{\mathcal{Y}} S_{\mathcal{Y}} \otimes U_{\mathcal{Y}}$$

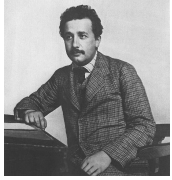
Young diagram



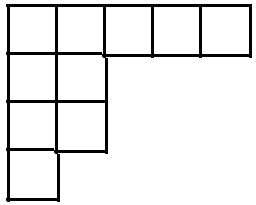
$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

Uses of



Young diagram

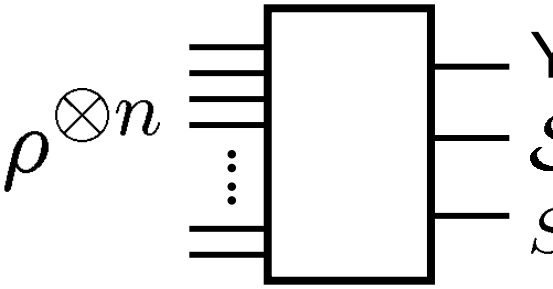


$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

What does this remind you of?

$\frac{k_i}{n}$ like probabilities like the spectrum of a density operator



Young diagram ← Optimal spectrum estimate

[Keyl and Werner. PRA, 2001]

$\mathcal{S}_n$ irrep space

$SU(d)$ irrep space

The Plethora of Uses of Schur



Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

Universal distortion-free entanglement concentration

[M. Hayashi and K. Matsumoto. Information Theory, 2004]

Universal compression with optimal overflow exponent

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Optimal quantum hypothesis testing

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Communicating without a shared reference frame

[S. Bartlett, T. Rudolph, and R. Spekkens, Phys. Rev. Lett., 2003]

Encoding into decoherence-free (noiseless) subsystems

[J. Kempe, D. Bacon, D. A. Lidar, and K. B. Whaley, Phys. Rev. A, 2001]

Simulating lattice gauge theories on a quantum computer

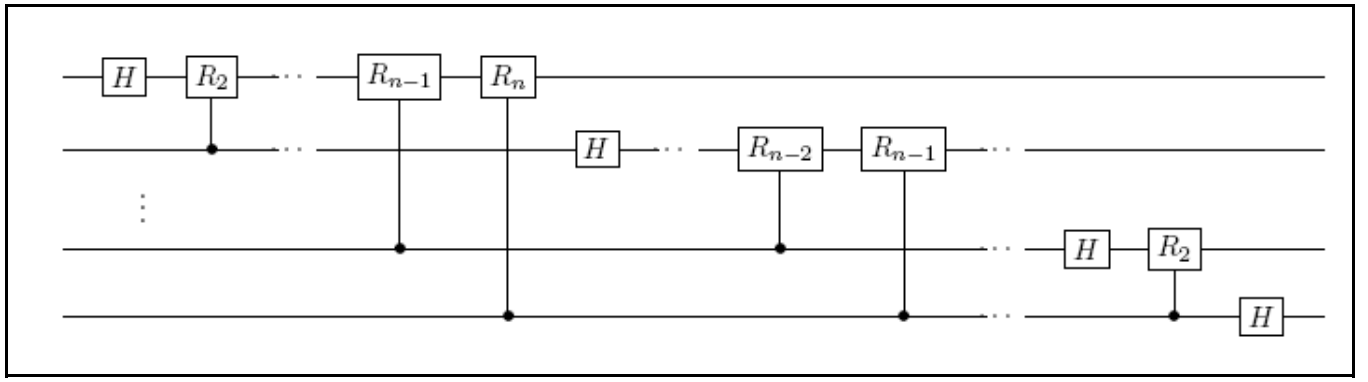
[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

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But Efficient?



QFT



Schu
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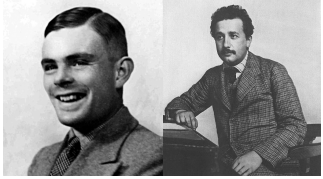


Who would need to efficiently calculate Clebsch-Gordan coefficients for SU(d)?



[L. C. Biedenharn and J. D. Louck, Commun. Math. Phys. 1968]

Efficient Schur



[D. Bacon, I.L. Chuang, A. Harrow, Phys. Rev. Lett. 2006]
[D. Bacon, I.L. Chuang, A. Harrow, SODA 2007]

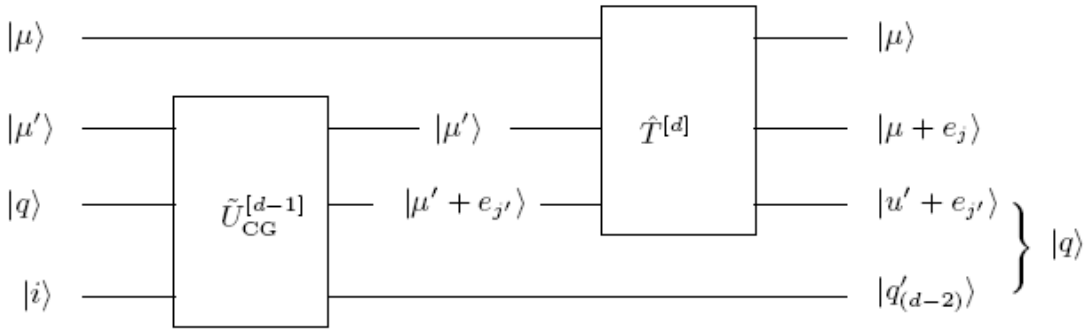


FIG. 4: The $\mathcal{U}_d$ CG transform, $\mathbf{U}_{\mathrm{CG}}^{[d]}$, is decomposed into a $\mathcal{U}_{d-1}$ CG transform $\tilde{U}_{\mathrm{CG}}^{[d-1]}$ (see Eq. (54)) and a reduced Wigner operator $\hat{T}^{[d]}$. In Fig. 5 we show how to reduce the reduced Wigner operator to a $d \times d$ matrix conditioned on μ and $\mu' + e_{j'}$.

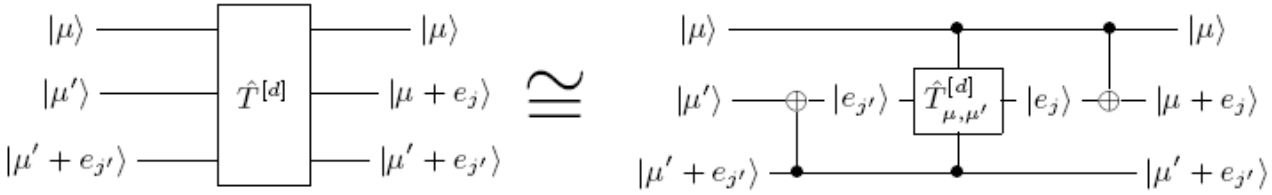


FIG. 5: The reduced Wigner transform $\hat{T}^{[d]}$ can be expressed as a $d \times d$ rotation whose coefficients are controlled by μ and $\mu' + e_{j'}$.

Schur transform to accuracy ϵ : time $poly\left(log\frac{1}{\epsilon}, n, d\right)$

The Plethora of Uses of Schur



Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

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[M. Hayashi and K. Matsumoto. Information Theory, 2004]

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But Algorithms?

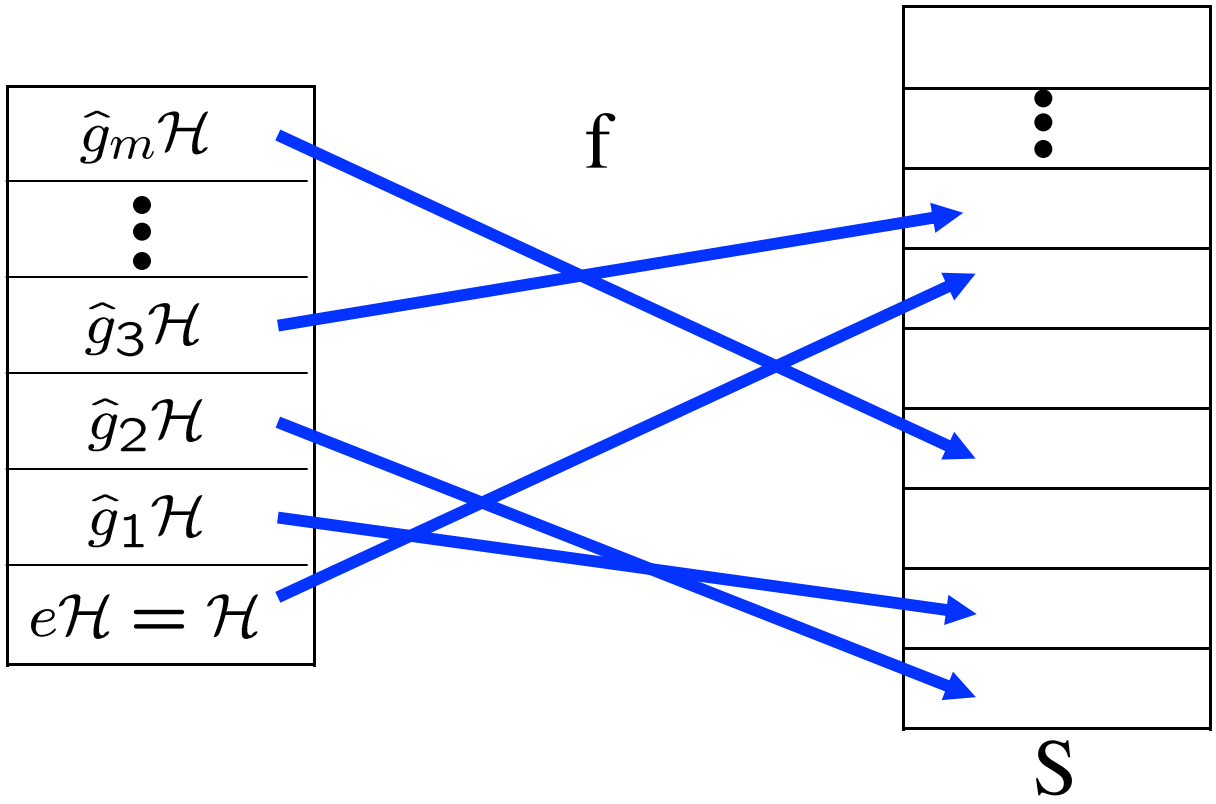
The Hidden Subgroup Problem



Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.

Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

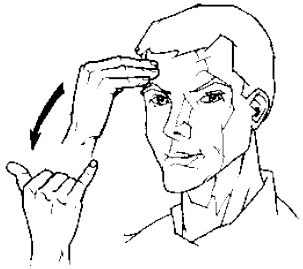
Efficient: If runs in time polylog($|\mathcal{G}|$)



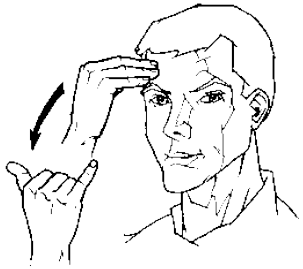
Hidden Subgroup Disease

“The quantum measurement problem refers to a set of people.”
- Hideo Mabuchi

“The hidden subgroup problem refers to a subgroup of people.”



Why the Fuss?





Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

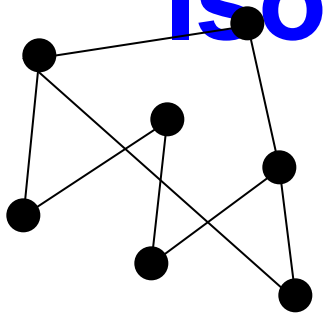
Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

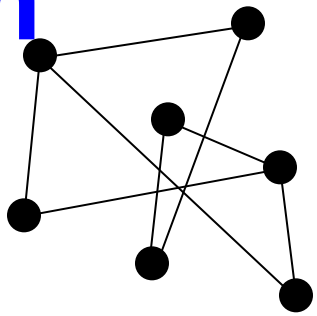
efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

Graph Isomorphism





adjacency matrix A_1



adjacency matrix A_2

$G_1 = (V_1, E_1)$ $G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

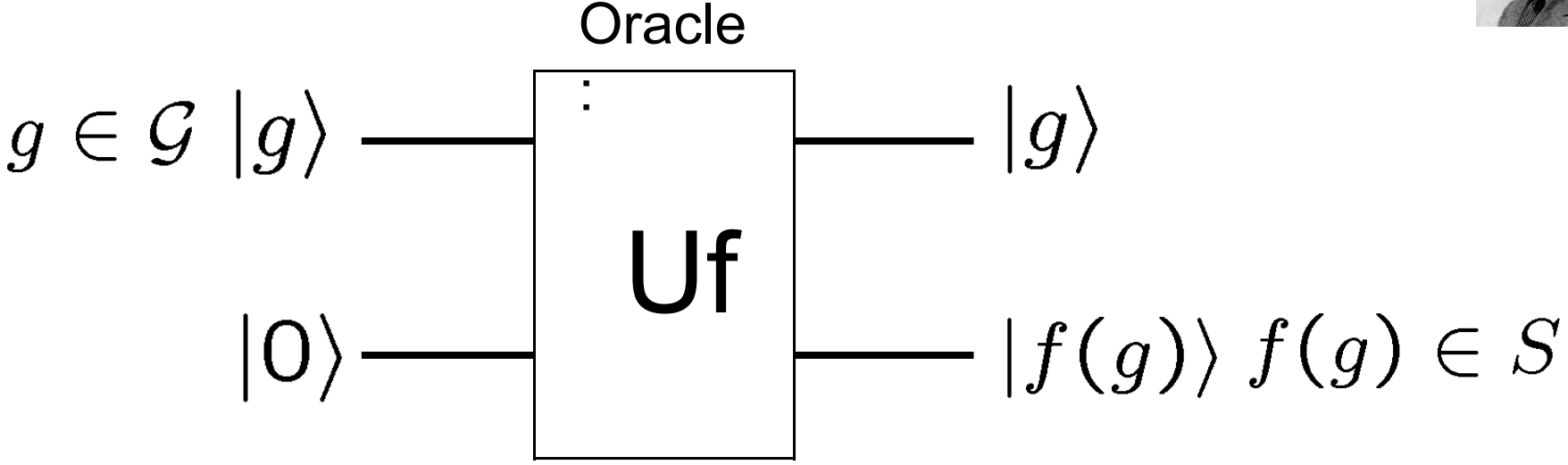
$$f : \mathcal{S}_{2n} \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f(\pi) = \pi(A_1 \oplus A_2)$$

If graphs are not isomorphic, hidden subgroup will not mix the graphs. If they are isomorphic, they will mix the graphs.

Standard Query Model





Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$ hidden subgroup state

You Down With Symmetry?

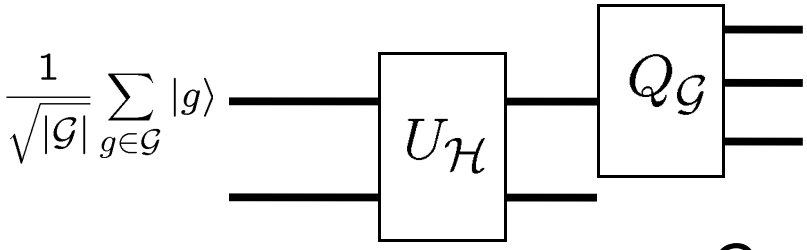


$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

Fourier transform over finite group $\mathcal{G}$

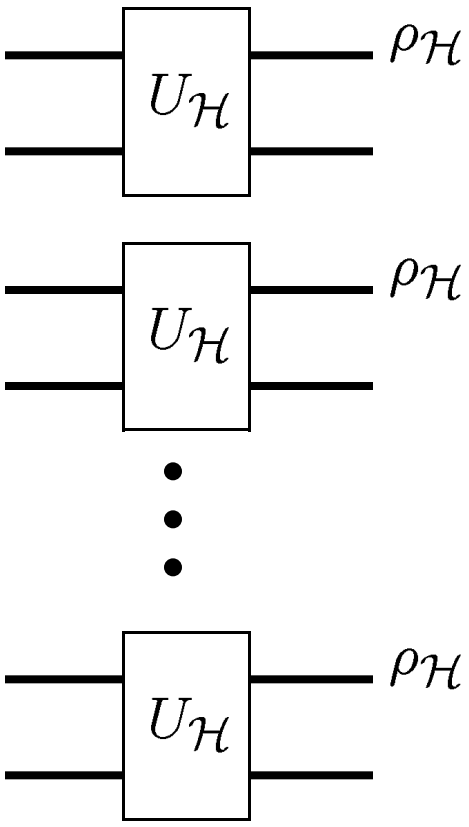


Can we solve HSP this way?

Multicopy







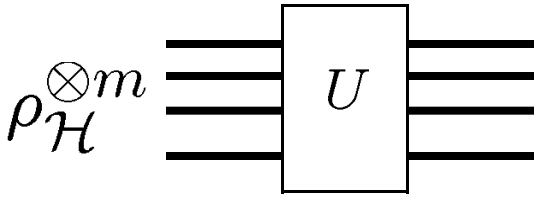
$\rho_{\mathcal{H}}^{\otimes m}$

Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) **NO**

[Moore, Russell, Hallgren, Roetteler, Sen 05]

Required:



You Down With Symmetry?



$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

Symmetry of two copies? $\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}}$

Permutation symmetry….measure the Young Diagram?

Nope [Childs, Harrow, Wocjan 06]

Heisenberg Group

Semidirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left(\begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

From the play “Copenhagen” Margrethe Bohr to Heisenberg:

“Your talent is for ski-ing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

Multicopy Success!



Efficient algorithm for the Heisenberg HSP: $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$

[D. Bacon, A.M. Childs, and W. van Dam FOCS 2005]

PGM/optimal measurement approach:

$$\rho_{\mathcal{H}} \qquad Pr(succ) = O\left(\frac{1}{p}\right)$$

$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}} \qquad Pr(succ) = O(1)$$

Efficient algorithm for the Heisenberg HSP:

1. PGM is optimal and succeed w.h.p.
2. How to efficiently implement the PGM?

But Why Does It Work?

Clebsch-Gordan Transform To The Rescue?

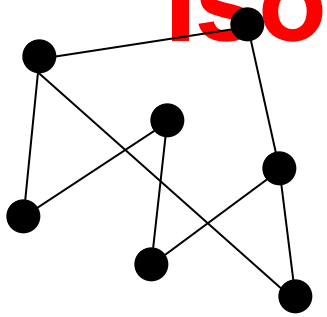


Problem:

$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}}$ does not have a Clebsch-Gordan related symmetry

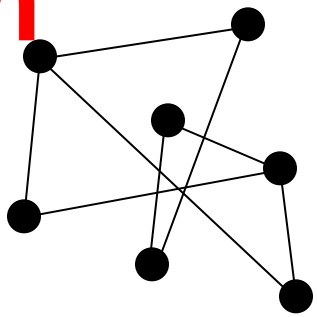


Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the V_2 vertices such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \oplus \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup has a nontrivial τ

Conjugate Subgroups

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$, are conjugate if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

This is an equivalence relation among the subgroups of $\mathcal{G}$

$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

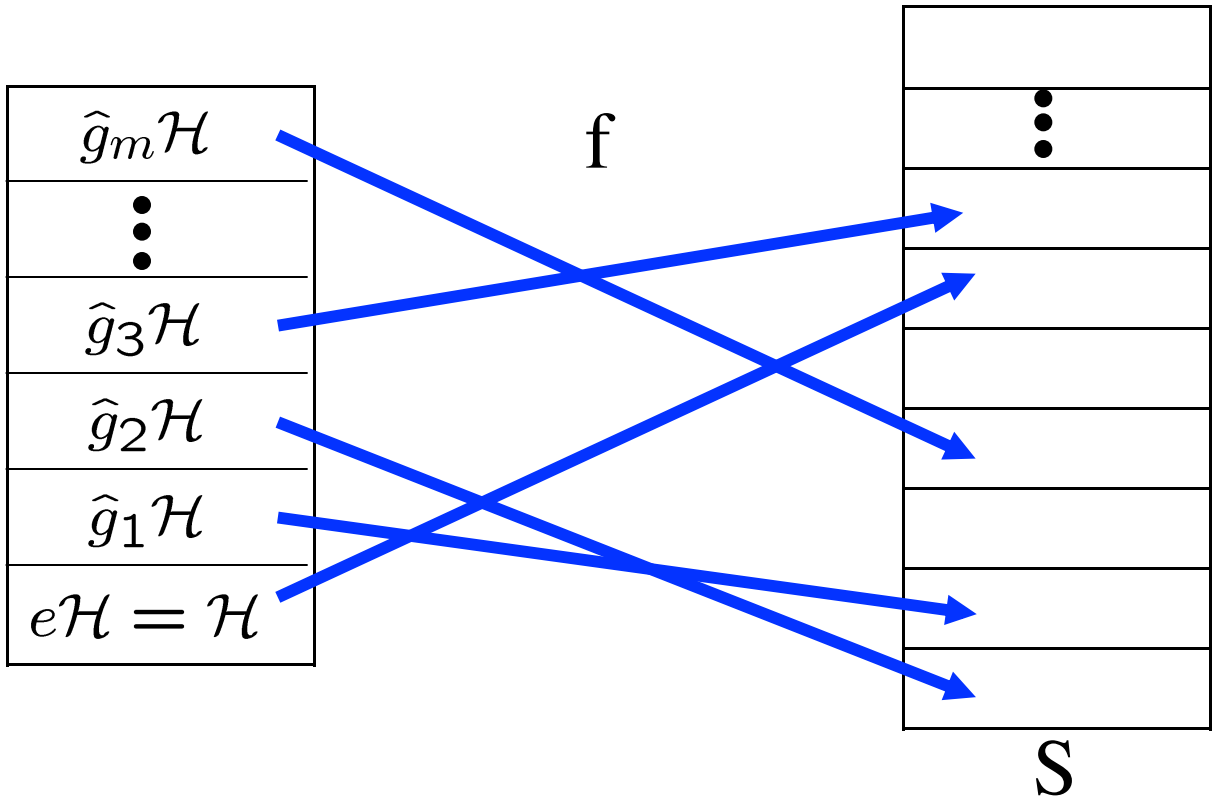
ir: voy, vas, va, vamos,

Hidden Subgroup Conjugacy Problem

Setup: Fix a group $\mathcal{G}$(known) and a subgroup $\mathcal{H} \subset \mathcal{G}$(unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

Problem: identify the conjugate subgroup class of $\mathcal{H}$.

Efficient: If runs in time polylog($|\mathcal{G}|$)



Relationships

How do the hidden subgroup problem (HSP) and the hidden subgroup conjugacy problem (HSCP) differ?

1. If group is abelian, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

2. If subgroups are normal, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

These are two cases where we know quantum computers can efficiently solve the HSP. Coincidence?

Symmetries of HSCP States

Hidden subgroup conjugate state

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}}$$

Hidden subgroup conjugate states

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

In context of optimal measurement method (PGM)
[e.g. Ip 03, Bacon, Childs, and van Dam 05]
this translates into extra symmetry of the optimal measurement

$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

Symmetries of HSCP States

What is the extra symmetry of the multiple HSCP states?

Clebsch-Gordan transform for the regular representation of finite groups:

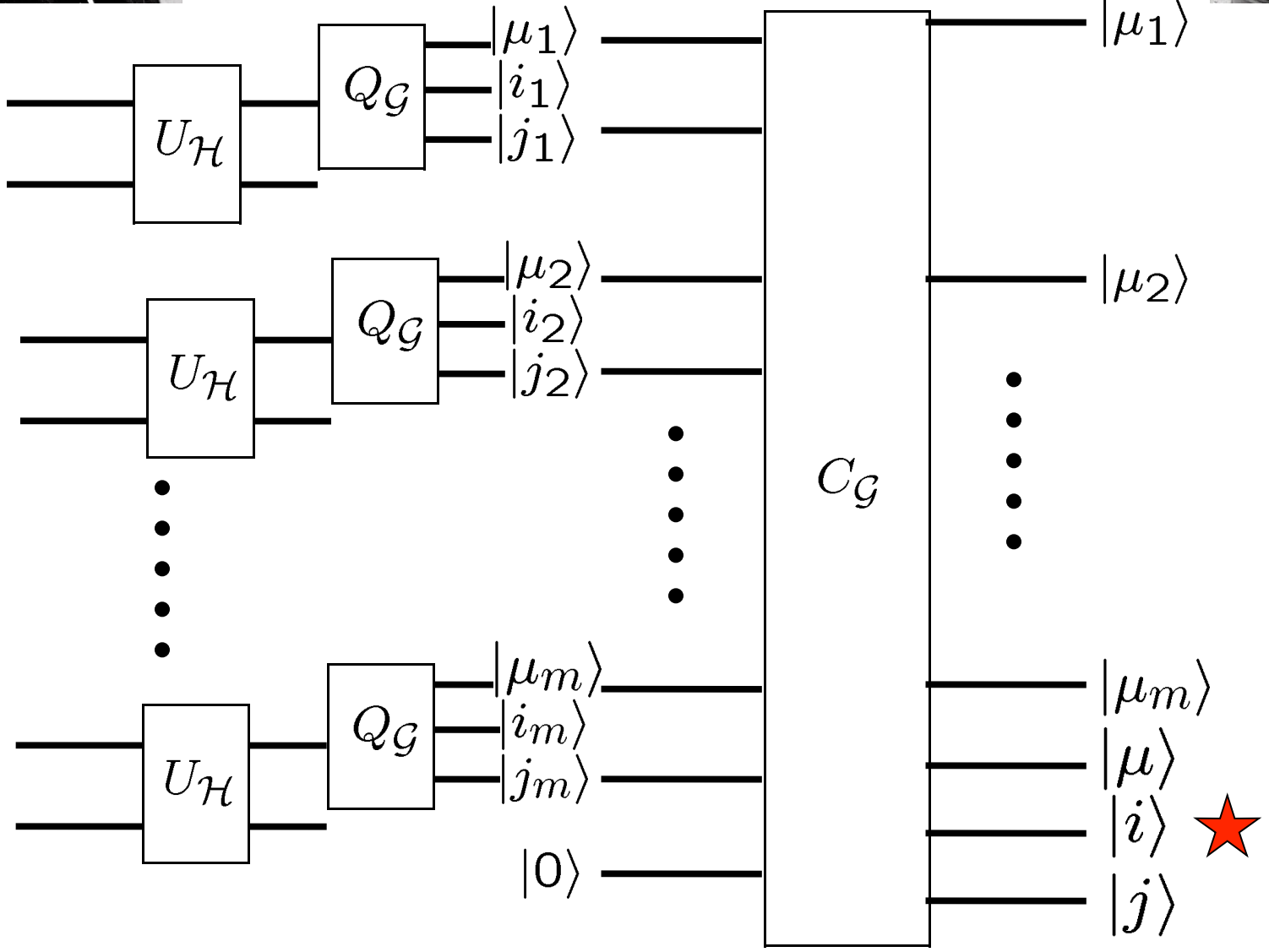
$$D_R(g)^{\otimes m} = \left[\bigoplus_{\mu}(I_{d_\mu} \otimes D_\mu(g))\right]^{\otimes m} \quad D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g) = \bigoplus_{\mu} \left(I_{n_\mu(\mu_1,\ldots,\mu_m)} \otimes D_\mu(g)\right)$$



General Circuit

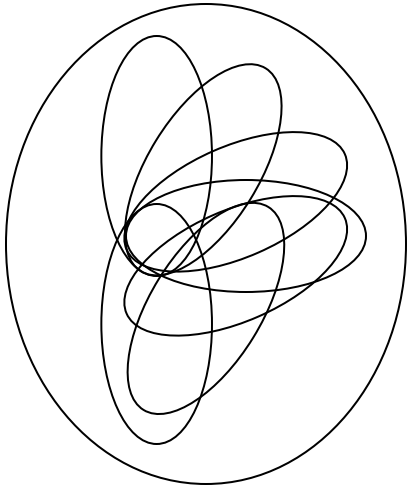




Heisenberg HSCP and HSP

[Bacon, arXiv:quant-ph/0612107]

Heisenberg HSCP ⇒ Heisenberg HSP



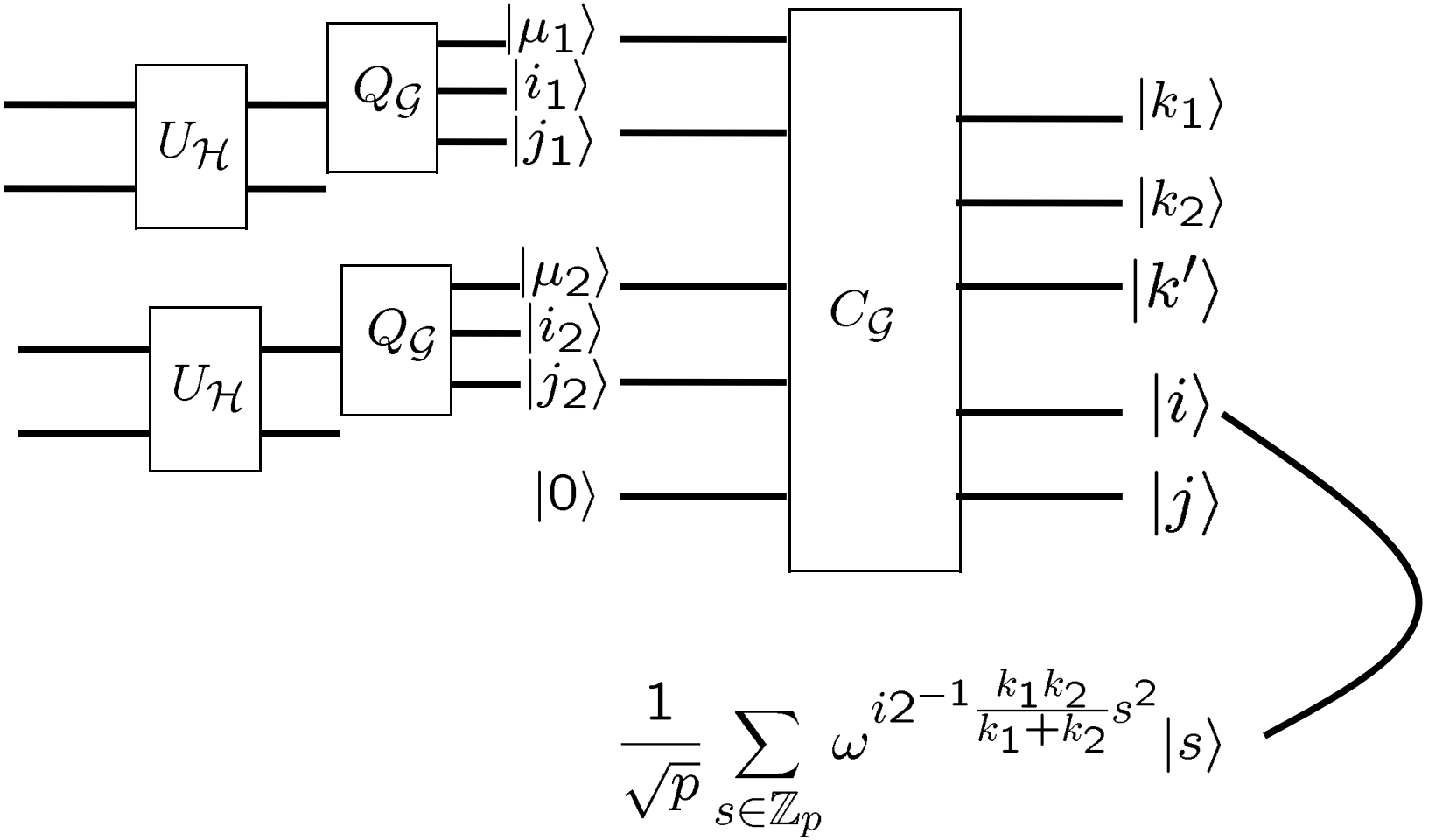
Conjugate subgroups are all normal subgroups of a subgroup of Heisenberg group

Identifying which conjugate subgroup identifies this subgroup

Thus solving HSCP implies solving HSP (use normal subgroup algorithm)

Heisenberg HSCP, CG Way

[Bacon, arXiv:quant-ph/0612107]



i labels set of conjugate subgroups

Heisenberg HSCP, CG Way

[Bacon, arXiv:quant-ph/0612107]

$$\frac{1}{\sqrt{p}} \sum_{s \in \mathbb{Z}_p} \omega^{i 2^{-1} \frac{k_1 k_2}{k_1 + k_2} s^2} |s\rangle$$

i can be found by performing

$$|t^2\rangle \rightarrow \frac{1}{\sqrt{2}}(|t\rangle + |-t\rangle)$$

and performing a
QFT

Clebsch-Gordan transform over Heisenberg group is related to “symmetry” which allows for efficient quantum algorithm for the Heisenberg HSP.

Heisenberg HS State

Restricted subgroups:

$$\mathcal{H}_{d_1,d_2} = \left\{ \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right), \left(\begin{pmatrix} 2d_1 + d_2 \\ 2d_2 \end{pmatrix}, 2 \right), \ldots \right\}$$

Random coset states:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$M^{(b)} = \sum_{i=0}^{b-1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^i = \begin{bmatrix} b & \frac{b(b-1)}{2} \\ 0 & b \end{bmatrix}$$

Coset labels: $l_1, l_2 \in_U \mathbb{Z}_p$

Hidden Subgroup states:

$$\rho_{d_1,d_2} = \frac{1}{p^2} \sum_{l_1,l_2 \in \mathbb{Z}_p} |l_1, l_2, d_1, d_2\rangle\langle l_1, l_2, d_1, d_2|$$

HS

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| l_1 + bd_1 + \frac{b(1-b)}{2} d_2, l_2 + bd_2; b \right\rangle$$

Perform the QFT over $(\mathbb{Z}_p)^2$ to the first two registers:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p^3}} \sum_{b, x_1, y_1 \in \mathbb{Z}_p} \omega_p^{(l_1 + bd_1 + \frac{b(1-b)}{2} d_2) x_1 + (l_2 + bd_2) y_1} |x_1, y_1, b\rangle$$

Measure these two registers:

$$x_1, y_1 \in_U \mathbb{Z}_p$$

$$|x_1, y_1, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \omega_p^{(bd_1 + \frac{b(1-b)}{2} d_2) x_1 + bd_2 y_1} |b\rangle$$

HS

Two copies: $x_1, y_1, x_2, y_2 \in_U \mathbb{Z}_p$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{(b_1 x_1 + b_2 x_2) d_1 + \left(\frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2 \right) d_2} |b_1, b_2\rangle$$

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle$$

Compute v and w:

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle |v, w\rangle$$

Uncompute

If we could erase the b1,b2 values:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Then a QFT over $(\mathbb{Z}_p)^2$ would find d1 and d2.

Remember v and w are a function of b1 and b2.

Solving Equations

Compute v and w:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle$$

$$v = b_1x_1 + b_2x_2$$

$$w = \frac{b_1(1-b_1)}{2}x_1 + b_1y_1 + \frac{b_2(1-b_2)}{2}x_2 + b_2y_2$$

For fixed x1,x2,y1,y2 let $S_{v,w}$ be the set of b1,b2

solutions to these equations

$$\frac{1}{p}\sum_{v,w\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}\left(\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle\right)|v,w\rangle$$

We want to implement $\frac{1}{\sqrt{|S_{v,w}|}}\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle|v,w\rangle \rightarrow |0,0\rangle|v,w\rangle$

Solving Equations

$$\frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle \rightarrow |0,0\rangle |v,w\rangle$$

Easier to think about the inverse (do it unitarily):

$$|0,0\rangle |v,w\rangle \rightarrow \frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle$$

i.e. given v and w, transform to superposition over solutions to the equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Assuming y1,y2,y1+y2 do not equal 0 (prob O(1/p)),
then

$\frac{p(p-1)}{2}$ solutions have $|S_{v,w}| = 0$

p solutions have $|S_{v,w}| = 1$

$\frac{p(p-1)}{2}$ solutions have $|S_{v,w}| = 2$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Focus on when there are two solutions.

We can use standard finite arithmetic to find these solutions. Use to create labeled superposition of solutions:

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |j, (b_1)_j, (b_2)_j\rangle$$

Hadamard first qubit, measure. If 0, you have

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |(b_1)_j, (b_2)_j\rangle$$

Heisenberg HSP Solved

This allows us to perform:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sqrt{|S_{v,w}|}\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Which has a O(1/2) fidelity squared with the state we desire.

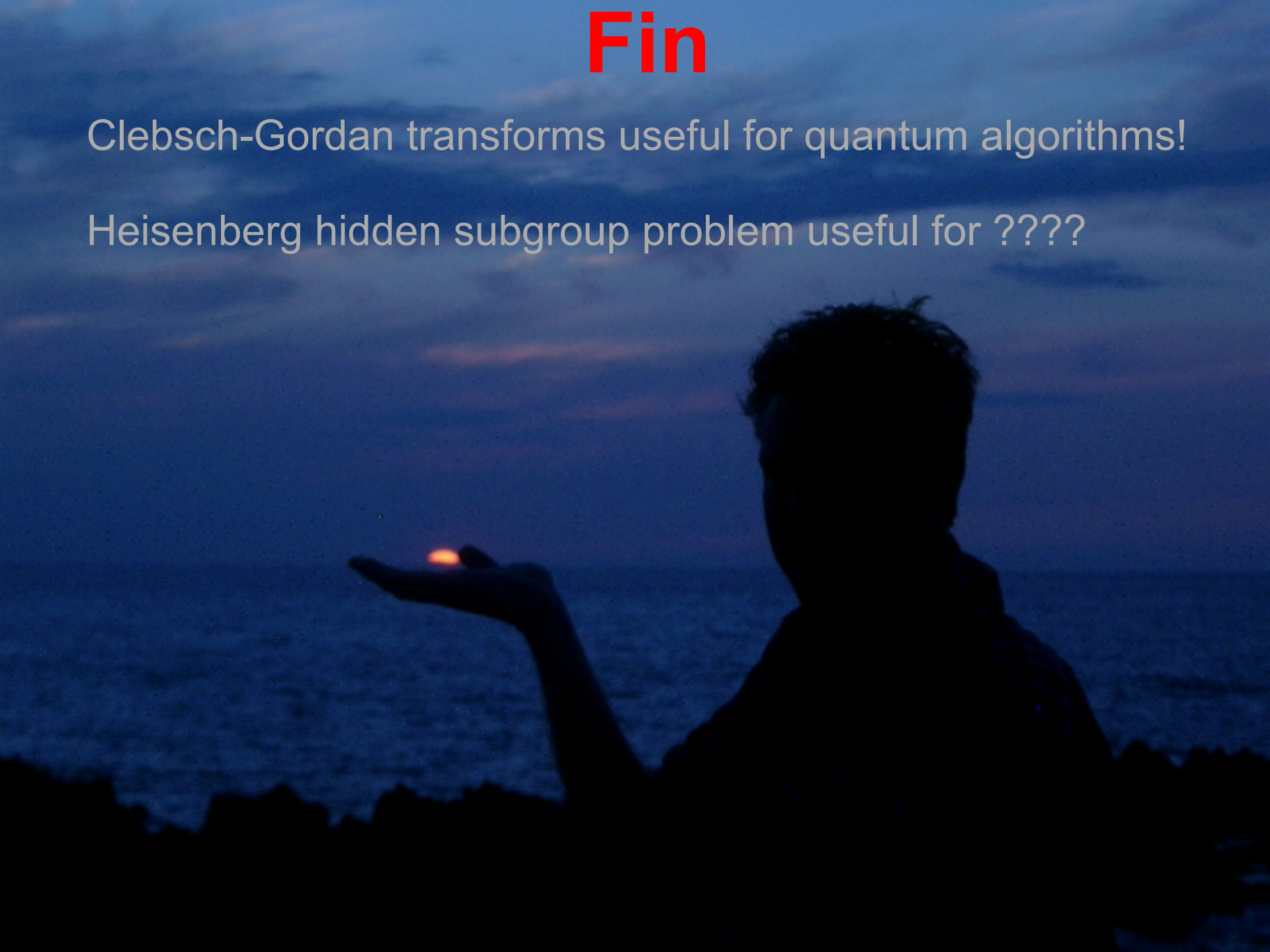
QFTing the second register will let us find d1 and d2

Efficient algorithm for the Heisenberg HSP!

Fin

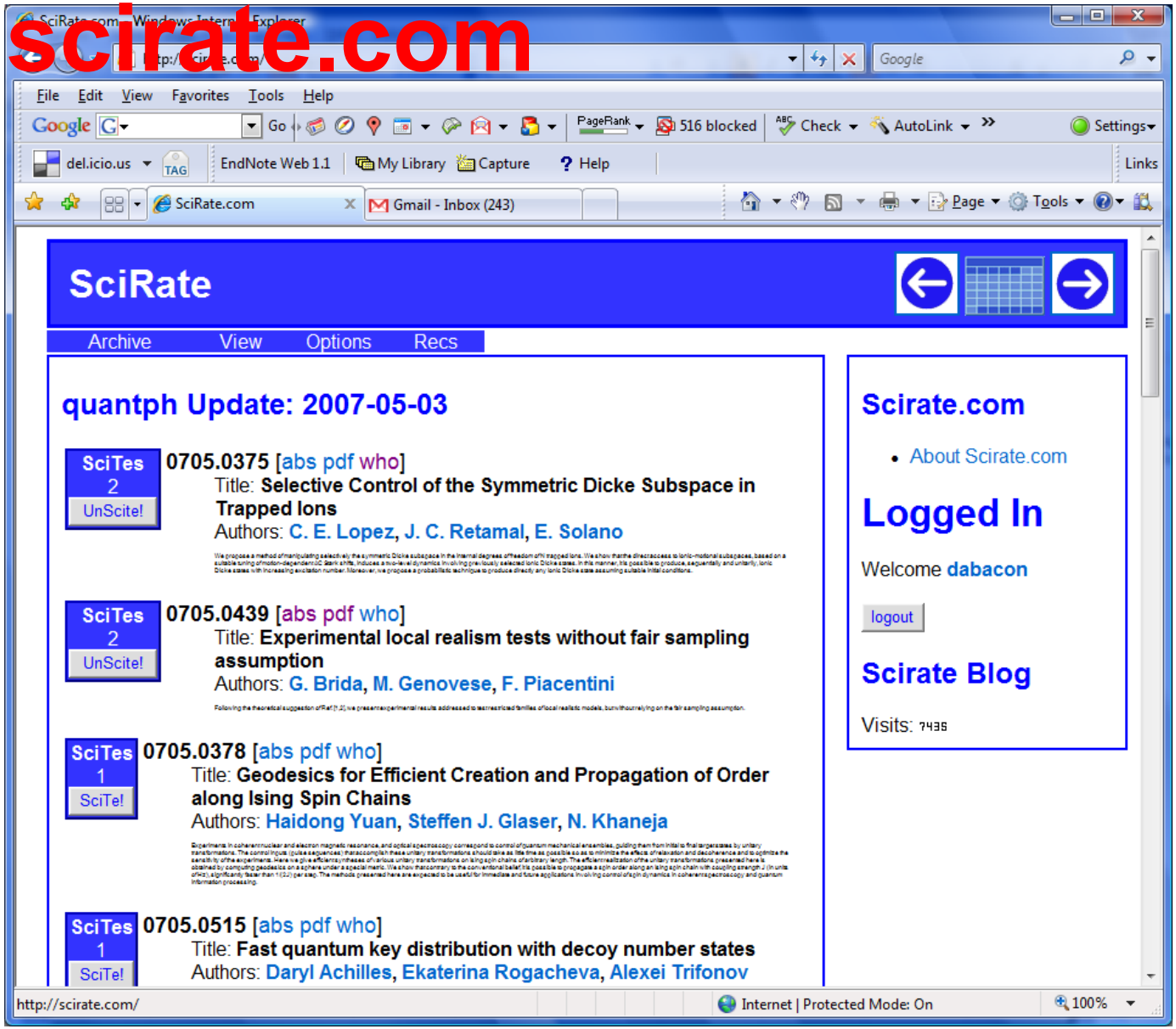
Clebsch-Gordan transforms useful for quantum algorithms!

Heisenberg hidden subgroup problem useful for ????



Quick Promo:

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Quantum Algorithms Using Clebsch-Gordan Transforms

Dave Bacon
Department of Computer Science & Engineering
University of Washington

funded by

DTO

NATIONAL SECURITY AGENCY
UNITED STATES OF AMERICA

NSF

SCIENCE TO SHAPE THE FUTURE OF THE ARMY
ARMY RESEARCH OFFICE
1951

|QW⟩ group

"Que Dubayah"



Collaborators: Childs [Caltech], van Dam [UCSB], Chuang [MIT], Harrow [Bristol]

HSP as State Distinction



Single Copy

Given: $\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Multiple Copy

Given: $\rho_H^{\otimes k}$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*

Holevo's Condition



Given: ρ_i with probability p_i

Measure: POVM with elements $\{G_j\}$

Outcome: j corresponds to guessing state ρ_j

Maximize: probability of successfully identifying ρ_j

Necessary and Sufficient Condition: [Holevo 73]

$$\sum_i p_i \rho_i G_i = \sum_i p_i G_i \rho_i \qquad \sum_i p_i \rho_i G_i \geq p_j \rho_j$$

- **Hard to find explicit forms for *Gj***

Pretty Good



Given: ρ_i with equal probabilities

Observation: the "Pretty Good Measurement" is often pretty good.

Pretty Good Measurement (PGM):

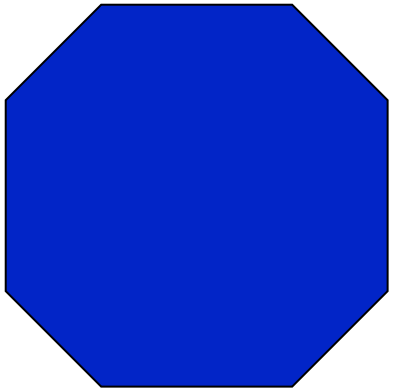
$$G_j = T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i \rho_i$$

(plus $I - \sum_j G_j$ if needed.)

A Strange Lock







Rotation : | 0 | 0 | 0 |

Reflection: | 1 |

Display: | BFD |

Setup: Display will give same result if we enact rotation by k and if we enact rotation by -k+x and a reflection.

x is the hidden combo.

Goal: Make display repeat to open the lock.

Dihedral Hidden Reflection Problem

Hidden Reflection



Given: $\rho_H^{\otimes k}$ with equal a priori probability

Dihedral HSP

$$H = \{e, rs^l\}, l \in \mathbb{Z}_N$$

Theorem: [Bacon, Childs, van Dam 05]

The PGM is the optimal measurement for identifying *H* (hidden reflection problem)

Theorem: [Moore, Russell 05]

The PGM is the optimal measurement for identifying *H* when *H* and *G* are a Gelfand pair.

arXiv generalization speed = Jan 10 – Jan 31 = 21 days

Success Probability



Probability of correctly guessing reflection:

$$p = \frac{1}{2^k N^{k+1}} \sum_{x \in \mathbb{Z}_N^k} \left(\sum_{t \in \mathbb{Z}_N} \sqrt{\eta_t^x} \right)^2$$

$\mathbb{Z}_N^k :=$ k dimensional vector of elements of $\mathbb{Z}_N$

$\eta_t^x :=$ number of subsets of *x* which sum to *t*

Two orders:

For *k<log N*, *p* is exponentially small (in log(N))

For *k>log N*, *p* is *O(1)*

transition at density = *d=1*

Must Work Hard

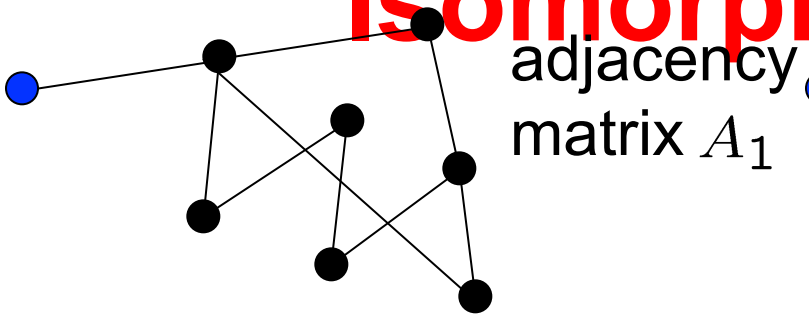


Theorem: [Bacon, Childs, van Dam 05]
Determining the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

Theorem: [Bacon, Childs, van Dam 05]
Determining the most significant bit of the hidden reflection requires *O(log(N))* copies of the hidden subgroup states.

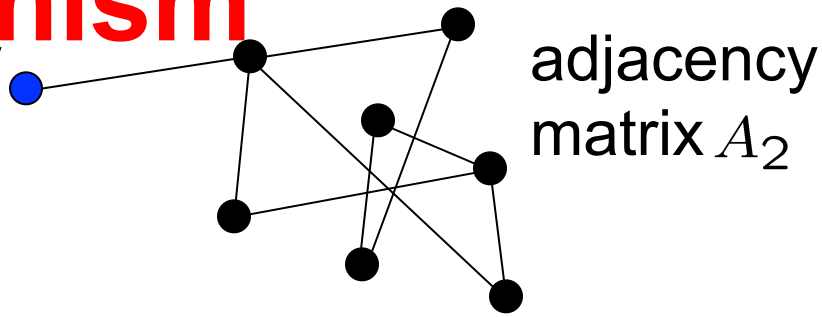
The Dihedral Hidden Subgroup problem looks hard!

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

If we want to find the isomorphism, we can attach extra vertices and edges to detect whether permutation of these vertices is in isomoprhism group.

HSCP → solve graph isomorphism and identify the isomorphism